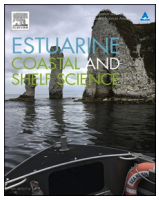
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Do you want your paper published? – Then referee others! The view from seasoned editors

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1. The peer review process and the associated problems

Journal peer review is the primary quality control mechanism in the scientific process of communicating results, which helps editors decide whether to publish a study in the form it was submitted by the authors to them, reject it, or recommend revisions before being accepted for publication (Malički and Mehmani, 2024). The peer review system has a long history in scientific publication. Csiszar (2016), provides some historical notes on the peer review consolidation: (i) in 1665, Henry Oldenburg, secretary of the Royal Society in London, created the Philosophical Transactions to simplify his correspondence, but he did not use a referee system; (ii) in 1699, the French Royal Academy of Sciences was given power by Louis XIV to report on and approve books for publication and bypass the royal censors, being the first example of peer review; (iii) in 1752, the Royal Society established a committee to vote on what to publish; (iv) in 1817, the geologist George Greenough introduced the term 'referee'; (v) in 1831, Cambridge professor William Whewell convinced the Royal Society to commission public reports on manuscripts; (vi) in 1833, the reports became private and anonymous; but (vii) it was not until the 20th century, when the system was fully established (e.g. it was as late as in 1973, when external refereeing became a requirement for publication in *Nature*).

Nevertheless, the process was *"troubled from the start"* (Csiszar, 2016), and some examples of such troubles are shown by this author and others (Joseph, 2024). The problems identified in these early times included lack of time of referees, bias in the process, inefficiencies in manuscripts handling, overburden of academics, etc. (Severin and Chataway, 2021).

The exponential increase in the number of manuscripts submitted in recent decades for publication is now overburdening the capability of available qualified referees to keep up with reviewing requests from editors and to ensure timeliness and quality of their evaluations (Stahel and Moore, 2014). Hence, in the 1970's (when *Nature* started to require external refereeing), the number of scientific papers published per year was <0.3 million, in 2000, it reached 2.5 million, and the most recent estimates indicate over 7 million per year (Fire and Guestrin, 2019). In fact, the number of papers published in 2022 was 47% higher than in 2016 (Abramo, 2024). This incredibly high number of papers has exacerbated the problems detected in the early times of the peer review process, and many editors (including us) have suffered the increasing difficulty of finding suitable referees and aggravating the delay and quality of peer review (Stahel and Moore, 2014; Barroga, 2020; Herculano-Houzel, 2024; Joseph, 2024; Mollaki, 2024). It is notable that the total time referees globally worked on peer reviews was over 100 million hours in 2020, equivalent to over 15 thousand years (Aczel et al., 2021), with an estimated monetary value of the unpaid time referees spent on reviews well over 2.5 billion USD in 2020.

Hence, in our and others' experience, there is an increasing unwillingness by scientists in accepting to peer review. Ben Messaoud et al. (2023) studied a total of 180,092 referees during the period 2018–2021, in 21 journals. They found a pre-pandemic rate of agreement in reviewing of 37.5%, which was maintained during the first phase of the COVID, and then decreased to 33% during the second phase of the COVID, as it was found by other recent quantitative studies (Künzli et al., 2022; Drozdz and Ladomery, 2024; Feetham-Walker et al., 2024). Some evidence in this increasing pattern is anecdotical (e.g Dance, 2023; Ellaway, 2024) and few others do not find evidence (Vines et al., 2010; Zupanc, 2024), but the number of referees per paper who need to be contacted by editors before being successful has significantly increased between 2016 and 2022 (Tropini et al., 2023). On a personal basis, both of us have been Editors-in-Chief since 2010 and have seen the number of referees approached often increased from 4 to 10–20.

Amongst many different reasons the referees declare to decline the invitation we can identify (Table 1): lack of time (by far the most important reason (Willis, 2016)), no academic or institutional recognition of this work, it is not paid work, too many invitations at the same time, the referee considers the journal or publisher predatory, outside of the expertise, etc. (Raniga, 2020). Most of these reasons are also

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Table 1
Some real acceptable and unacceptable reasons given by referees, invited by the authors of this manuscript as editors in: *Estuarine, Coastal and Shelf Science, Continental Shelf Research, Frontiers in Marine Science* and *Frontiers in Ocean Sustainability*.

Acceptable reasons	Unacceptable reasons
•Conflict of interest •Too busy •On sabbatical/parental leave •Too much teaching •Out of my area of expertise •I'm retired or on holiday •A poor paper, badly written which is wasting my time •Not understandable paper •I'm reviewing other papers now	•"I disagree with the publishing model" •"I dislike the publisher as they are making too much money and expecting us to do this for free" •"I won't referee non-native English speaker papers" •"I won't referee for certain countries for political reasons (e.g. Russia, Israel)" •"I won't referee for countries such as India and China as the papers are always poor" •"It is not a good use of my time" •"The head of department tells me I should spend time writing papers instead of reviewing them"

provided in a recent survey among 3,046 referees from 105 countries (Feetham-Walker et al., 2024). It is difficult to say how many declining invitations (or not response) we receive, but there appears to be only about 20–40% of invitations accepted and even this level declines at vacation times. In recent years, it is more common that editors need to invite more than 30 potential referees (one of us has a record of 72 invitations sent), before having at least two referees able to complete their reviews. Similar numbers (and reasons) are raised by other editors (e.g. Joseph, 2024). As editors, we would prefer to receive a response, and say you cannot do it, rather than not responding to our requests, since this blocks the system and produces unnecessary delays. It is true that there is also a problem created by the publishers, in not having referee details up to date, and by referees not updating them, so a large number of requests go unanswered or unable to pass institutional firewalls. We have also come across cases where a referee sits on a competing paper to one of their own to delay publication of the other; distasteful academic rivalries are unfortunate!

There is also a possibility that these peer review problems can have a 'Western-centric' academic perspective, that might differ across regions, languages (non-English speaking regions), institutions, level of development countries, or fields of study. The most larger study possibly is that from Ben Messaoud et al. (2023), with 180,092 referees, and supports this evidence, since 53.1% of referees were from Europe, 28.8% from North America, 7.9% from Asia, 7% from Oceania, 1.8% from Africa, and 1.3% from South America. These authors found that the agreement to review was highest among referees from Asia (52.3%), and then Africa (50.3%), South America (50.2%), North America (37.9%), Oceania (35.8%), and Europe (34.0%). More than 93% of referees were affiliated to a high-income country, whereas agreement to review was highest among referees affiliated to lower middle, low income, and upper middle income countries compared with high-income countries (60.2%, 54.8%, 51.8%, and 36.0%, respectively) (Ben Messaoud et al., 2023).

We take the view that for the peer-review system to continue, an author has to play their role. However, we have seen also that senior researchers (and especially those highly respected) do not accept reviewing in recent times. We realise that those researchers may have a high workload and other responsibilities, but they should be able spend some time helping and mentoring other researchers. Hence, perhaps we should invite more and more early career researchers. This would be good, since they can learn and understand the whole scientific process at an early stage, but the quality of the reviews could be lower, despite their good attitude. In turn, it is probably true that postdoctoral researchers, especially will be more up to date with the literature and methods than many senior scientists. It is also the case that seeing up-to-date research before publication will have a greater benefit, especially for such early career researchers.

Another problem is the editorial policy of many journals, which tends to reduce the time allowed for reviewing from four weeks, to three and now to one, emulating the timing for refereeing of high impact journals (e.g. Nature, Science). Of course, this discourages the potential referee – journals have to realise that scientists are not sitting idly by waiting for papers to review! We fully realise that many would like to referee papers, but the time is not suitable and they have too many personal and professional demands on their time.

Of course, we need to link the difficulties of peer reviewing to the current mixture of publishing models: subscription, hybrid and fully open-access, which can also discourage some referees. Hence, with the subscription model, the referee can accept to review. In turn, if an Open Access journal requires paying (and in some journals, such as those in the group *Nature*, you can pay over 10,000 euros), the referee can say that (s)he also wants to be paid. Yes, we accept that this has the unintended consequence that people can do quick, simple and bad reviews only to get the money, but an editor would have to decide if a review was satisfactory for the reward. We would be pleased to make such an evaluation of a review if it improves the refereeing process.

In the last two years, the potential use of artificial intelligence not only in writing papers, but also in peer reviewing them, has increased and led to academic concerns (Blau et al., 2024; Buriak et al., 2024; Saad et al., 2024). In a recent and inconclusive study by Feetham-Walker et al. (2024), among 3046 referees, 35% of respondents thought that generative artificial intelligence tools would have a negative impact on the peer review process. 36% were neutral, and 29% considered that it would have a positive impact. For those considering that these tools can provide useful outputs, most respondents think that expert human verification and editing is always required. Our minds are open to both, but under the control of the editor, author and the referee, so, using wisely it as a help to speed-up the process and improve the quality of the reviews. Artificial intelligence will increase the amount of plagiarism, as it only pulls information from that already published, including errors uploaded or even are created by on-line material. However, there will be value in ChatGPT e.g. in checking the language of the paper for non-native English speakers, or even in finding the most suitable referees for a paper. For achieving this, some authors, such as Mollaki (2024), advocate to go beyond policies by creating transparent procedures that will enable journals to investigate allegations of non-compliance and take decisions that will protect the integrity of the peer review system.

1.1. *Are there solutions?*

A little while ago, in discussing the same problem as here, a popular science magazine suggested that the peer-review system is in danger of collapsing through overweight, and then the only alternative is to allow anyone to upload a paper to a web site but then any comments, good or bad, could be added to the paper and made public so that eventually authors would be more careful. We doubt whether this would encourage quality and probably allow any rubbish to continue to be uploaded!

We emphasise that any solution proposed should be adopted, and accepted, by the publishers, journals, editors, referees and authors, thereby building a peer reviewing culture that ensures that all parties feel appreciated, otherwise, the problems will remain (Herculano-Houzel, 2024). Below we propose some solutions.

- If you wish to publish for a subscription publisher (not in a specific journal), X papers in a year, you should be obliged to review at least the same number of papers, if the journals are under subscription. If an author consistently declines, without acceptable reasons, they should not be able to publish in that outlet.
- Given that each paper needs 2–3 referees and as some papers are rejected, then an author should referee 4–5 papers for each one they

submit for the system to function. In fact, Aczel et al.(2021) found that on average, a referee completes 4.73 reviews per year.

- If you are invited by a journal which is hybrid or open access, you should be compensated, either with credits allowing you to publish for free, after several invitations (this is the case in *Frontiers* or *MDPI*) or paid an amount (e.g. *The American Economic Review* had paid referees $100 for timely reviews, *The Lancet* offers small honoraria for fast-track reviews). In our proposal, the boundaries could be 5000 euros as APC (author publishing charge): if the publishing costs are lower, then you receive credits, if the costs are higher, then you should be paid. Considering the mean time for review (established in 6 h) and the value of review labour, established by Aczel et al. (2021), this payment should be between $200 and $400 per review, depending on the countries, and this solution is gaining supporters (Seghier, 2024). However, some editors dislike this solution (Joseph, 2024), since it can increase the fraud in peer reviewing, incentivizing poor-quality reviews (Stahel and Moore, 2014). We argue that if, even with many more customers, airlines can have frequent flier systems and major stores can have loyalty schemes then publishers should be able to do the same – the publishers have to remember that the referees are needed for maintaining quality.
- Journals already offer certificates of reviewing, if needed, and it is becoming more standardized, allowing the authors to justify the number of reviews completed as an academic merit (e.g. the *Web of Science Researcher Profiles* let researchers track, verify, and showcase their peer review and editorial activities, raising academic profiles and credibility). Many journals even have the possibility of rating the quality of the revision work done by referees, and the editors must complete these rating to reward referees. We note that in some national research evaluation schemes, such as the UK, refereeing counts as an 'esteem' factor in the evaluation metrics.
- Editors and senior scientists as part of their duties must encourage and mentor early career scientists to review papers, formally training them by editing and journal platforms, which should invest part of their benefits in this task.
- Referees should remember that publishers have statistics on their performance so editors can see who are good and bad at responding – time taken, decision reached, number accepted, and to make sure they are not overloaded but also see who never responds – very often it is the busy person who responds positively (following the adage 'give a job to a busy person'!). Indeed, in platforms such as Editorial Manager, when searching for a referee a warning comes up as 'beware of using this referee as they have declined x times in the past year'! However, we acknowledge that some declines are not the fault of the referee who might not receive the invitation as it was treated by spam by the firewall of their institution. Hence, journals/publishers must help by keeping their database up to date – check email addresses and remove those which bounce back but authors have to address such firewalls that may treat review requests as spam, etc.
- Editors and senior scientists need to emphasise the benefits of refereeing (Table 2) for both the author and the referee. We must make people realise that they do have a duty to referee papers, and we repeatedly need to mention the alternatives (if no one referees then there is no quality control on the papers). If it remains difficult finding referees, then we have a spiral of decline - those who agree have more work and editors start getting downhearted (and probably give up!).

Finally, we emphasise that peer-reviewing is like democracy – it may be a flawed system, but it is the best we have at the moment. You should treat a paper for reviewing exactly as you would want your own paper treated!

Table 2
Benefits of reviewing for editors, authors and referees.

Benefits of reviewing for the editor/publisher:	Benefits of reviewing for the author:	Benefits of reviewing for the referee:
• Maintaining the quality of the journals, • Detection of plagiarism, • Helping the editor with papers, especially those outside his/her own field • Helping the editor to take the best decision on publishing a paper • It may reduce the costs for the journal as otherwise someone else would have to do this currently unpaid work	• Having a better paper, • Undertake a better future research, • The prevention of errors, • An improved reputation, • A greater networking capability	• Most referees see it as their "sense of duty" to review, to give back to the scientific community • More experienced referees see the value of mentoring young researchers • Enjoyment in reviewing • General interest in the area • Awareness of new research and developments before their peers • Further career development • Help with own research or new ideas • Association with journals and Editors • Keep updated with latest developments

CRediT authorship contribution statement

Angel Borja: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Michael Elliott:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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No data was used for the research described in the article.

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